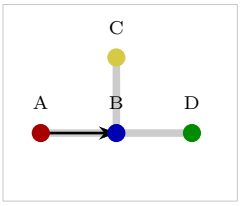
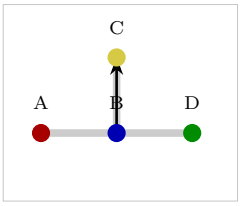
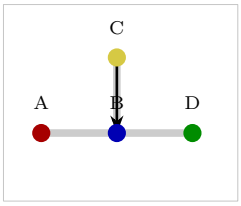
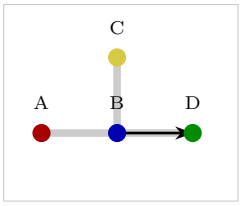
Move a pawn on the roads. Keep each journey as ordered step tiles. Two adjacent steps along the same road in opposite directions are an out-and-back pair. Remove only out-and-back pairs, and replay the other steps in order. The map, start, and finish stay fixed.

input



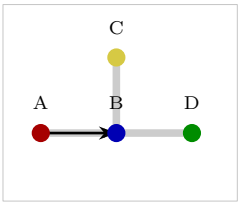


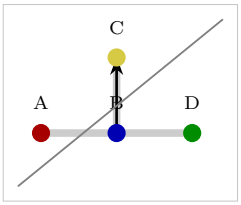


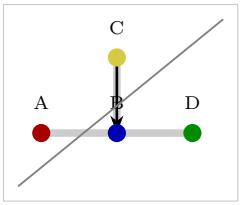


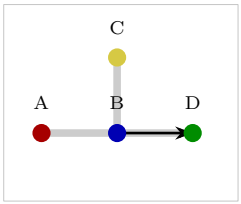
$A \to B \to C \to B \to D$

remove

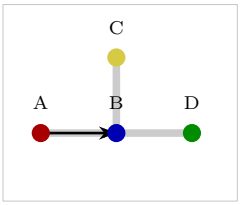


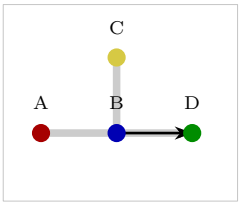






output

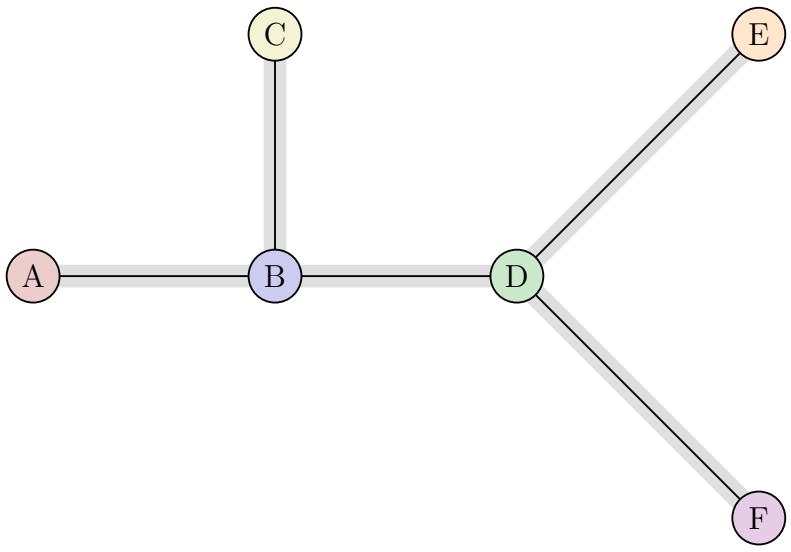




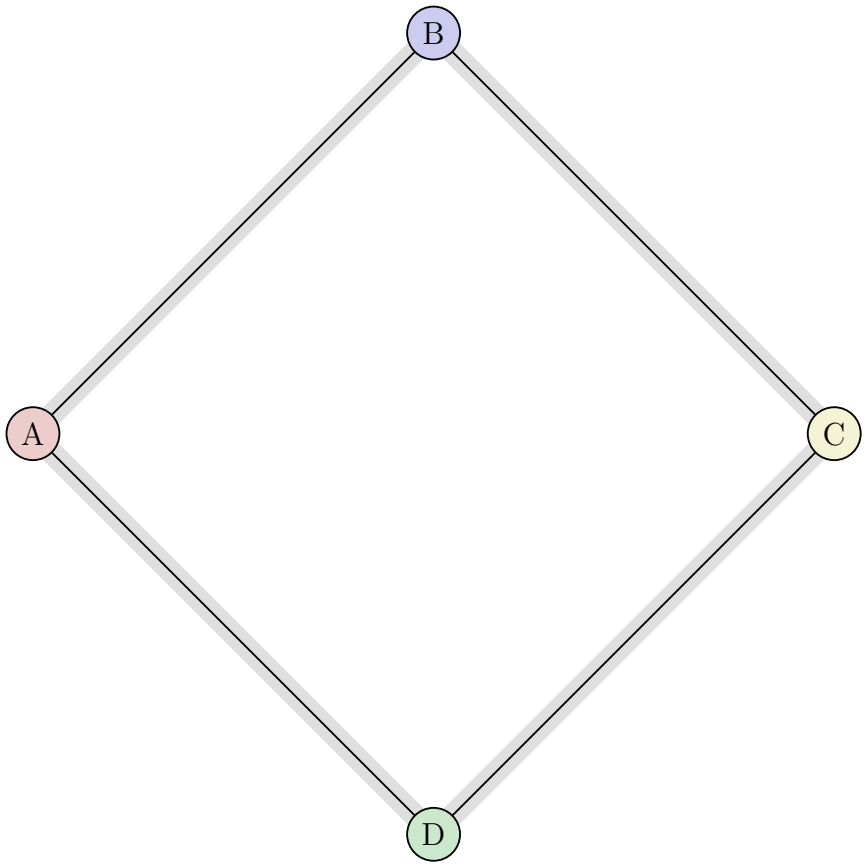
$A \to B \to D$

No steps left means the pawn stays at its start.

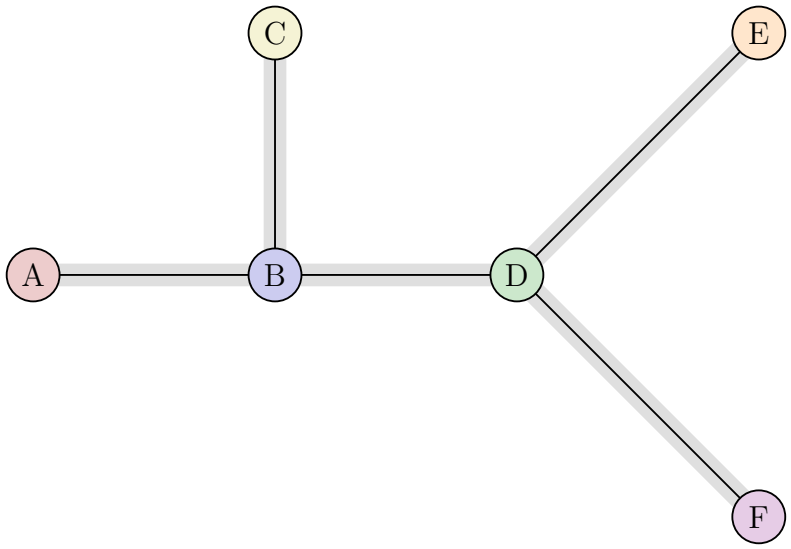
Problem 1: Make a three-step trip from A to E and four-step trips from B back to B and from A to D. Make each one as a row of step tiles, then shorten it as much as you can.



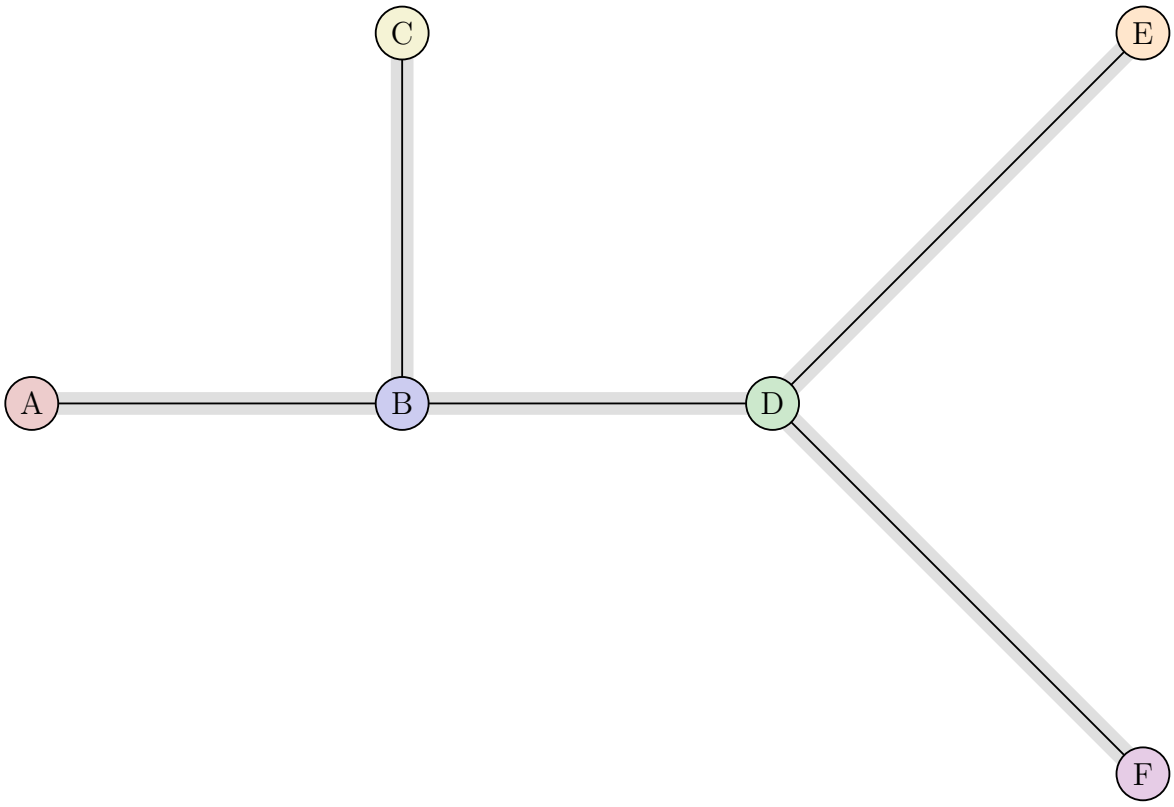
Problem 2: Make four-step trips from A back to A on this ring. Which can you shorten to staying at A, and which keep some steps?



Problem 3: Make a seven-step trip from A to E and a seven-step trip from C to F. Replay each one, removing out-and-back pairs until none remain.

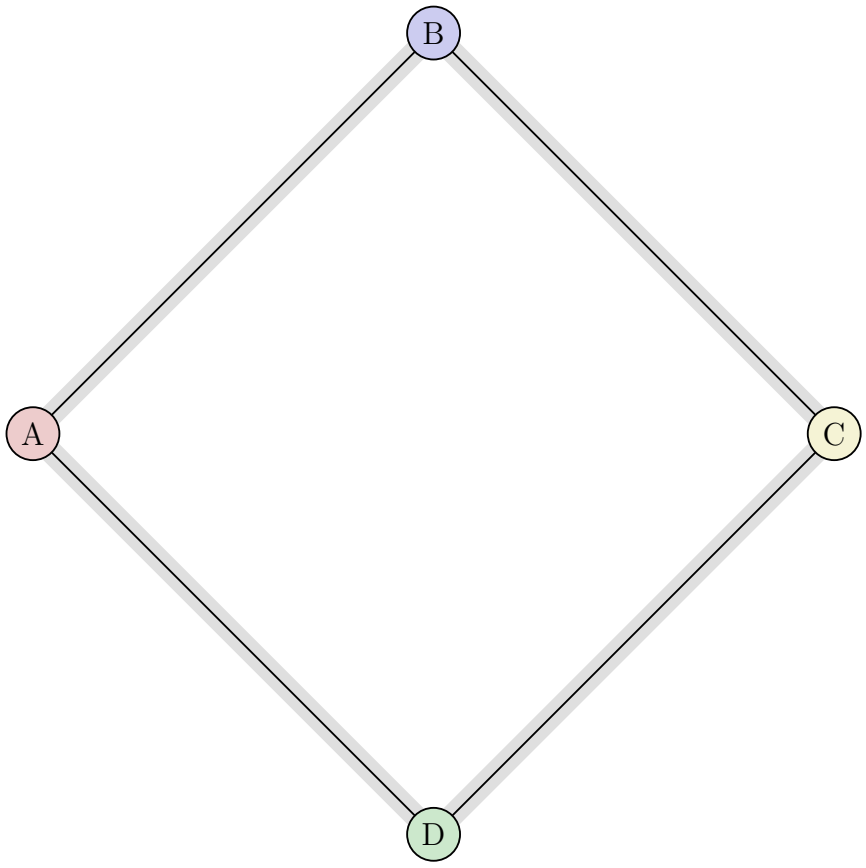


Problem 4: Make trips from A back to A using 4, 6, 8, and 12 road steps. Which can you shorten to staying at A?



Problem 5: Make trips from C to F that use every road and shorten each one. Can two of them end differently?

Problem 6: Make eight-step trips from A back to A and shorten each. Find every different result.



Problem 7: Make every different trip from A to C that can remain after shortening a six-step trip. Can two of your remaining trips use the same four roads?

